

INNOVA JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION II
in preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE
NAME

CLASS

INDEX NUMBER

BIOLOGY

9648/02

Paper 2 Core Paper

17 September 2013

2 hours

Additional Materials: Answer Paper
Cover Page

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer **any one** question.

At the end of the examination, fasten all your work securely together.

The number of marks is given in the brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	10
2	14
3	16
4	14
5	15
6	11
Section B	
7 / 8	20
Total	100
Percentage	
Grade	

This document consists of **18** printed pages.



Section A
Answer **all** questions.

- 1 Fig. 1 is an electron micrograph of a cell found in the pancreas.

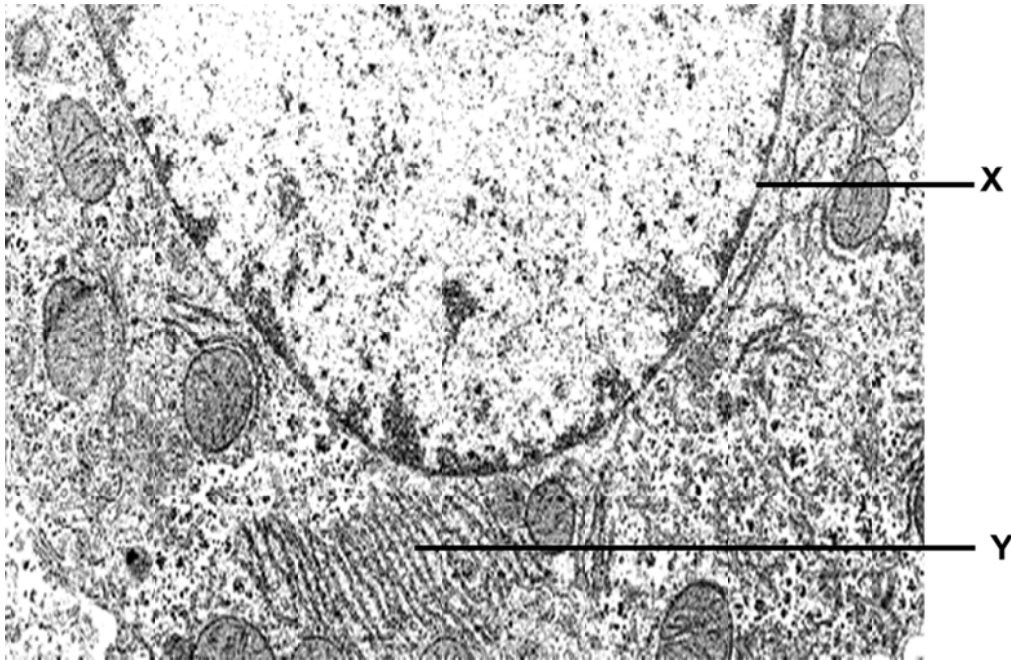


Fig. 1

- (a) Describe how organelle X is adapted for genetic expression.
1. *Nuclear pores with proteins controlling movement of RNA and proteins between nucleus and cytoplasm;; (MUST)*
 2. *Nuclear envelope separates nuclear compartment from cytoplasm (in eukaryotes) which allows spatial separation of the transcription and translation process → creates micro-environment that allows for post-transcriptional level of control / modification of transcripts before translation;;*
 3. *Contain genes which code for products involved in genetic expression (eg. HATs);;*
 4. *Contain enzymes e.g. HATs that chemically modify chromatin, affecting compactness and accessibility of promoter to transcription factors and RNA polymerase, hence allowing for genomic level of control of gene expression;; (KIV 3-4)*
- [2]

(b) Describe how insulin, after being synthesized by ribosomes on Y, is exported out of the cell.

1. *protein travels thru cisternae to bud off in a transport vesicle at transitional ER, transport vesicle travels through cytoplasm & fuse with GA at cis face;;*

2. *protein enters cisternal space & modified by adding/deleting/substituting of sugar monomers in oligosaccharide chains &/or adding phosphate groups;; ® post-translational mod*

3. *modified & tagged protein travels from cisternae to cisternae via budding & fusion of Golgi vesicle, sorted protein leaves GA in a secretory vesicle that buds off at trans face;;*

4. *secretory vesicle moves near cell surface membrane via cytoskeleton, membrane of secretory vesicle fuses with plasma membrane → insulin released into exterior of cell via exocytosis;;* [4]

(c) When insulin is released into the bloodstream, it exerts many effects on carbohydrate, lipid and protein metabolism.

One of its effects is to activate the enzyme glycogen synthase, which functions to polymerize glucose monomers into glycogen.

(i) Explain how the structure of glycogen enables it to perform its function in animals.

<i>Structure of glycogen</i>	<i>Function</i>
<i>1. Composed of several hundred / thousands of glucose molecules as monomers which are linked by α glycosidic bonds</i>	<i>1. Stores large amount of energy in glucose molecules i.e. main <u>respiratory substrate</u>, α-1,4 glycosidic bonds can be hydrolyzed by enzymes present in plants & animals to release glucose molecules for respiration;</i>
<i>2. Highly branched</i>	<i>2. Easily hydrolyzed to <u>by enzymes</u> to release glucose monomers when energy is needed;</i>
<i>3. Large molecule, high molecular weight</i>	<i>3. Kept within cell, cannot diffuse out of cell; insoluble, therefore osmotically inactive;</i>
<i>4. -OH groups not protruding outwards;</i>	<i>4. Does not interact with water, insoluble, osmotically inactive, can be stored without causing influx of water;</i>

Any 2, 1 mark each

[2]

- (ii) Describe **two** ways in which the structure of glycogen differs from the structural polysaccharide found in plant cell walls.

Glycogen	Cellulose
1. Made up of α glucose monomers;	1. Made up of β glucose monomers;
2. α -1,4 and α -1,6 glycosidic bonds link glucose monomers together;	2. β -1,4 glycosidic bonds link glucose monomers together;
3. highly branched structure;	3. long, unbranched linear chains;
4. -OH groups not protruding outwards of macromolecule/ face interior of macromolecule;	4. -OH groups projecting out of cellulose chains to allow cross-linkages to be formed b/w OH groups of parallel cellulose chains;
5. α glucose monomers not flipped 180° with respect to previous monomer;	5. β glucose monomers flipped 180° with respect to previous monomer;

Any 2, 1 mark each

[2]

[Total: 10]

- 2 Fig. 2.1 shows the structure of a protease enzyme found in a retrovirus.

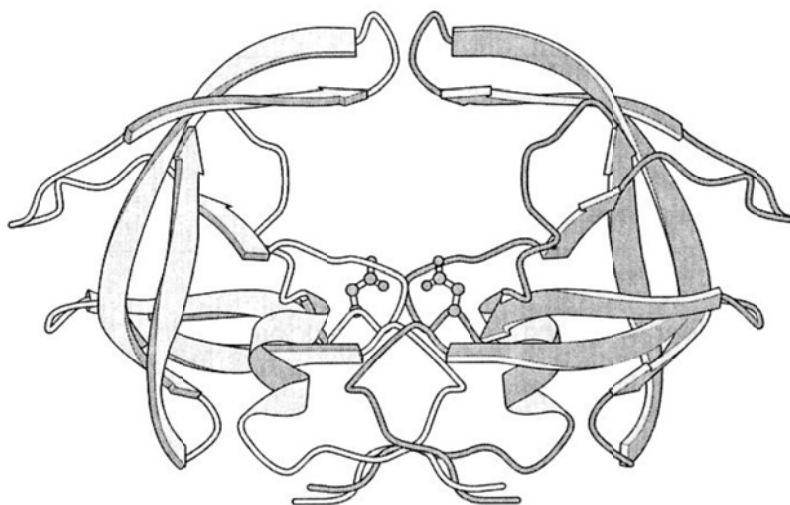


Fig 2.1

- (a) (i) With reference to Fig. 2.1, state the evidences that the enzyme has both a tertiary and quaternary protein structure.

1. Folding of polypeptide chain to bring α -helices and β -pleated sheets into specific 3-D structure (via intramolecular bonds) $\rightarrow$ shows tertiary structure;;
2. Enzyme made up of 2 polypeptide subunits held together by intermolecular bonds $\rightarrow$ shows quaternary structure;;

Accept " $3^\circ/4^\circ$ " but circle

[2]

- (ii) Explain what determines the 3-dimensional configuration of the enzyme.

1. Protein structure maintained by bonds between R groups of amino acids in polypeptide chain;;

2. Hydrophobic interactions between non-polar R groups, Hydrogen bonds between R groups of polar amino acids, Ionic bonds between oppositely charged R groups, diS bonds between SH groups;; (Any TWO) [2]

- (b) The rate of reaction of HIV protease may be decreased by different inhibitors.

Fig. 2.2 shows the effect of substrate concentration on the initial rate of an enzyme catalysed reaction with and without the addition of a non-competitive inhibitor.

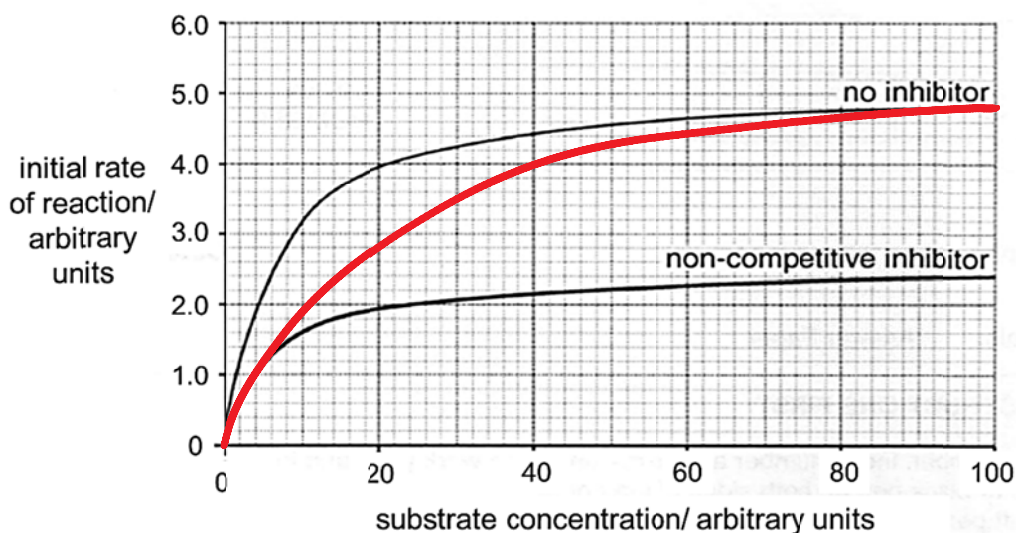


Fig 2.2

- (i) On Fig. 2.2 draw and label the curve you would expect in the presence of a competitive inhibitor. [1]

curve with initial rate < w/o I that reaches V_{max};;

- (ii) Describe and explain the difference in the shape of the curve you have drawn and that of the non-competitive inhibitor.

1. Quote V_{max} (and K_m) for both graphs;;

**2. At low [S],
Competitive inhibitor has similar structural shape to S/ close structural resemblance
VS
Non-competitive inhibitor is structurally different from S, binds at site away from enz active site
BOTH → inhibit S binding / E-S complex formation thus decreasing rate of rxn;;**

3. **At high [S],**
Competitive inhibition: higher chances of S binding to the active site outcompeting inhibitor hence effect of inhibitor becomes negligible / WTE
VS
non-competitive inhibition: effective enz concentration reduced;; [3]

- (c) A similar protease is found in blackflies. A proteinaceous toxin found in insecticides used to control its population can be activated by this protease.

The active toxin binds to receptor molecules on the larvae's gut epithelial cell membranes. This damages the epithelial cells and kills the larvae.

Fig. 2.3 shows the results of an experiment to determine the optimum pH for activation of the toxin.

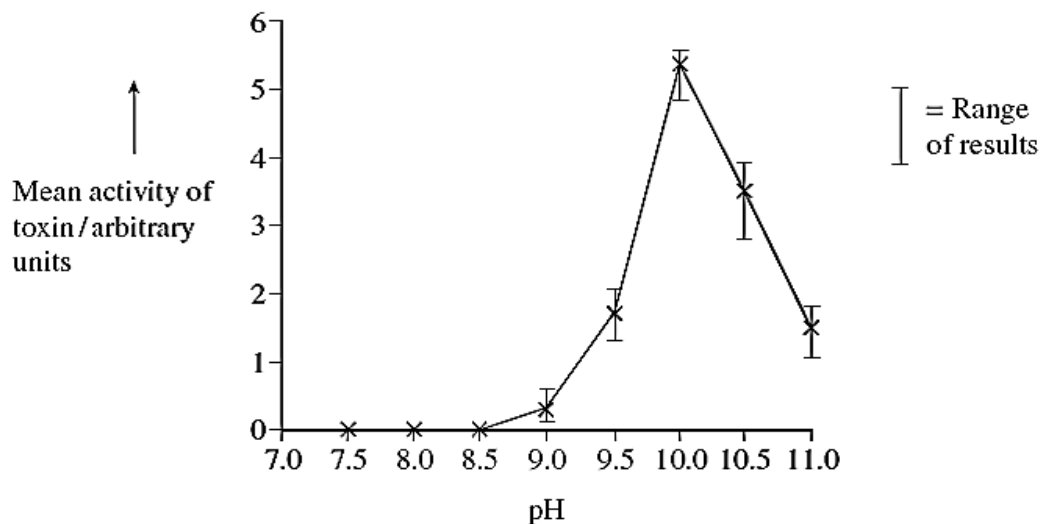


Fig. 2.3

- (i) With reference to Fig. 2.3, describe the effect of pH on mean activity of the toxin.
- Optimum pH at 10 at which there's highest mean activity of toxin between 5-6 a.u (specify one value);;**
 - either an increase in pH from 10 to 11 or decrease in pH from 10 to 9.5 results in drastic decrease of mean activity to 2 a.u (or less);;**
 - further decrease in pH from 9.5 to 8.5 results in mean activity decreasing from 2 a.u to 0 a.u**
- [2]

- (ii) Explain why there is no activity of toxin when pH falls below 8.5.
1. *increase H^+ concentration disrupts intramolecular ionic bonds and hydrogen bonds that help maintain the specific 3D conformation of the protease;; ® toxin*
 2. *Disruption of secondary and tertiary structures/ denaturation of protease → loss of specific 3D conformation of enz active site → toxin (substrate) cannot bind to active site of protease → no activation of toxin;;* [2]
- (iii) Suggest how the protease in larva's digestive system causes the toxin to be activated.
1. *protease cleaves toxin when catalytic aa at its active site hydrolyses peptide bonds in toxin;;*
 2. *toxin molecule acquires complementary configuration for binding receptor molecule/ exposes binding site for receptor molecule on gut epithelial cell membrane;;* [2]

[Total: 14]

- 3 Fig. 3.1 shows a T4 bacteriophage which has been modified to contain the DNA of Mu bacteriophage, a known temperate phage.

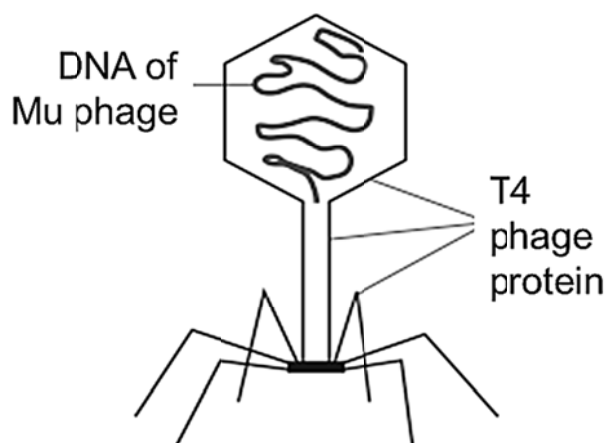


Fig. 3.1

- (a) Suggest how this bacteriophage attaches to the correct host cell.
1. *Long tail fibres recognises specific receptors on host cell surface membrane and bind via weak interactions;;*
 2. *(Short tail fibres)/ base plate pins interact strongly to anchor phage to host;;* [2]
- (b) Outline the main events which occur following the entry of the phage DNA into the host cell until new phage particles are released.
1. *Mu phage genome integrates into host cell chromosome → prophage → replicated together with host cell chromosome;;*
 2. *Prophage increase in number of copies until exposure to certain chemicals/ UV, prophage induced to exit bacterial chromosome;;*

3. Prophage transcribed into structural proteins (capsid proteins, head and tail/ tail fibre structures) using host cell RNA pol and ribosomes → these together with viral genome assemble to form complete Mu viral particle;;

4. phage directs the production of an enzyme that breaks down the bacterial cell wall from within fluid enters the bacteria, causing it to swell and undergo cell lysis → release new Mu phage particles;;

® everything if T4 is released

[4]

While the T4 bacteriophage infects bacterial cells, the influenza virus infects cells of the respiratory tract in animals.

H1N1 influenza virus causes swine flu in pigs. In a number of instances, people have developed the swine flu infection when they are closely associated with pigs, for example farmers and pork processors.

Figure 3.2 shows a H1N1 virus.

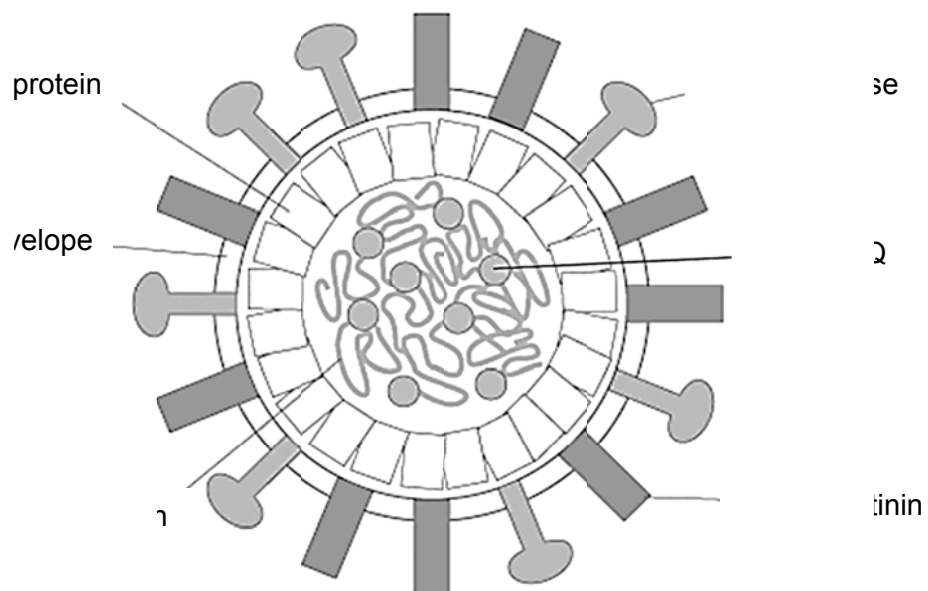


Fig. 3.2

(c) Describe the significance of enzyme Q to the reproductive cycle of the H1N1 virus.

1. RNA-dependent RNA polymerase uses viral ssRNA as template for transcribing complementary RNA strands;;

2. which are used as (1) templates for transcription of new RNA strands which form viral genome, and (2) used as mRNA for translation of viral proteins;;

3. eg. capsid proteins/ glycoproteins/ enzymes → enable viral assembly into new progeny viruses;;

[3]

(d) A person is infected with the H1N1 influenza virus but does not develop the symptoms. Examination shows that the person's immune system has produced an antibody that binds to the haemagglutinin molecules.

(i) Suggest why the person does not develop influenza-like symptoms.

1. antibodies produced bind to HA preventing them from recognizing and binding to (complementary) sialic acid residues on cell surface receptors on epithelial cells/ antibody-coated/ tagged viruses are recognized by phagocytes;;

2. virus is unable to enter the cells to cause cellular damage/ engulfed and destroyed by phagocytes before viral entry occurs;;

[2]

(ii) After a few days and many viral generations, the person begins to develop influenza symptoms. Tests for the antibody referred to above show that it is still present in the blood.

Account for the observations and comment on the implication this has for developing H1N1 influenza vaccines.

1. virus mutated due to the lack of proofreading by the viral RNA pol → structurally different HA that is not complementary to the antibody;;

2. Need for regularly designing new/ different vaccines;;

[2]

(e) Contrast the reproductive cycles of the H1N1 influenza virus and Human Immunodeficiency Virus (HIV).

	H1N1 Influenza Virus	HIV
1. Host cell	Epithelial cells of respiratory tract	CD4 T-lymphocytes
2. Attachment	HA attaches to sialic-acid containing residues on cell surface receptors of host cell → RME	Gp120 binds to CD4 receptors on T-lymphocytes, undergoes conformational change to expose fusion domain of gp41 → membrane fusion
3. Penetration	Acidification of endosome via M2 ion channel → conformational change of HA → release viral capsid into host cytoplasm, capsid proteins removed to release viral RNA	Viral capsid released into host cytoplasm, capsid proteins removed to release viral RNA
4. Integration of viral genome	No integration of viral genome into host chrom	Viral genome/ DNA integrated into host cell chrom via integrase
5. Reverse transcription of viral genome	No reverse transcription of viral genome/ ssRNA used as template by RNA pol to synthesize new viral genome and mRNA for translation into new viral proteins	Reverse transcription of ssRNA → dsDNA via RTase

(Any 3, 1 mark each)

[3]

[Total: 16]

- 4 Oxytocin (OT) is a neuropeptide secreted from the posterior pituitary gland in the brain and is best known for its role in mammalian birth and lactation.

In response to a variety of stimuli such as suckling, parturition, or certain kinds of stress, the processed OT peptide is released from the posterior pituitary into the systemic circulation.

The OT receptor's activity is mediated by G proteins which activate a phosphatidylinositol-calcium second messenger system.

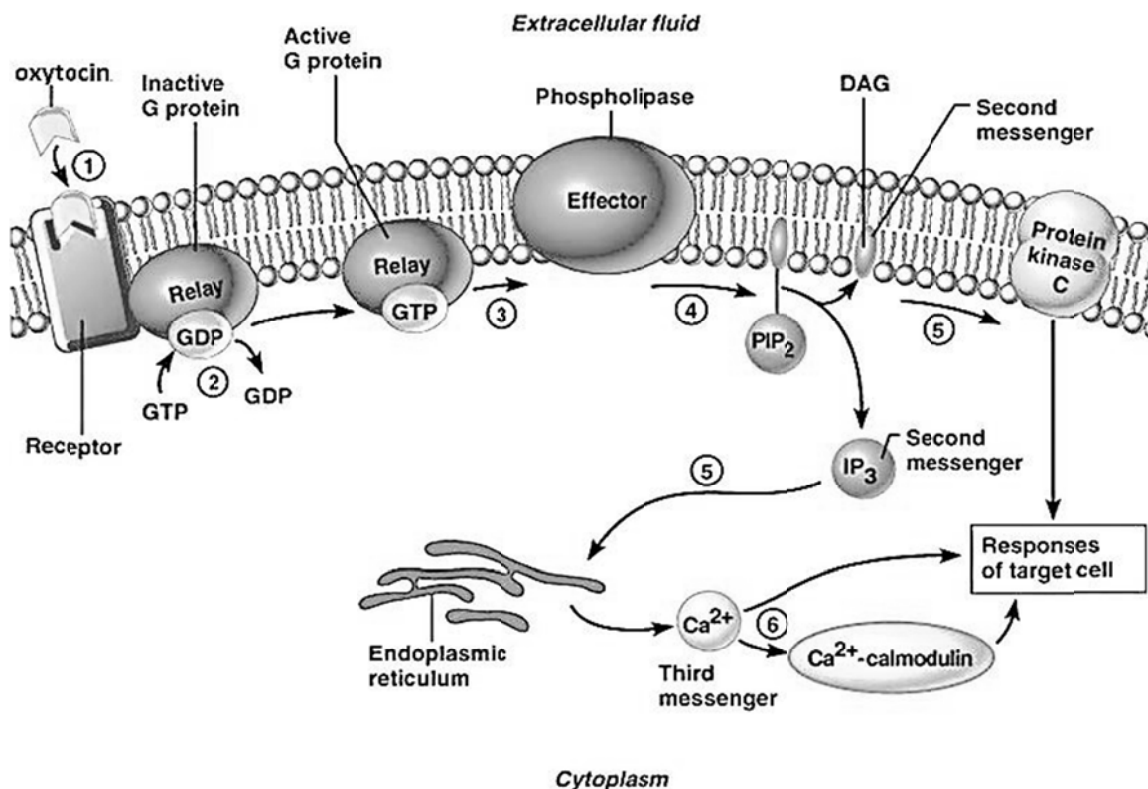


Fig. 4.1

- (a) The blood brain barrier is set by the endothelial cells lining the vessels in the brain.

Suggest why these cells can prevent oxytocin secreted from the gland to re-enter the brain.

1. CSM of cells made up of lipid bilayer with hydrophobic core made up of hydrocarbon tail of fatty acids;;

2. Oxytocin is peptide hormone which is hydrophilic → unable to traverse lipid bilayer/ too large/ absence of transmembrane transport protein for oxytocin;;

[2]

(b) With reference to Fig. 4.1, describe and explain how a molecule of oxytocin can lead to signal amplification leading to cellular responses.

1. **Oxytocin binds to GPCR (at extracellular face), which has an associated inactive G-protein → change in conformation of GPCR;;**
2. **stimulates GDP exchange for GTP at α -subunit of G-protein → translocated along cytoplasmic face of CSM towards effector/ phospholipase;;**
3. **phospholipase cleaves many PIP_2 molecules into second messengers DAG and IP_3 / signal amplification at this stage;;**
4. **DAG activates protein kinase C which carries out target cellular responses;;**
5. **IP_3 stimulates the release of Ca^{2+} ions from (s)ER, Ca^{2+} forms complex with calmodulin which carries out target cellular responses;;**

Reference to signal cascade

***mech of action: literature seems to imply 1 DAG → 1 PKC** [5]

γ -Aminobutyric acid (GABA) is the chief inhibitory neurotransmitter in the mammalian central nervous system involving mature neurons. During foetal development, the release of GABA leads to an excitatory post-synaptic potential in immature neurons.

During labour, findings shown that high levels of maternal oxytocin induced a switch in the action of GABA from excitatory to inhibitory in the neurons of the foetus at birth hence increases their resistance to damage during delivery.

Fig. 4.2 shows the generation of the post-synaptic potentials in immature and mature neurons.

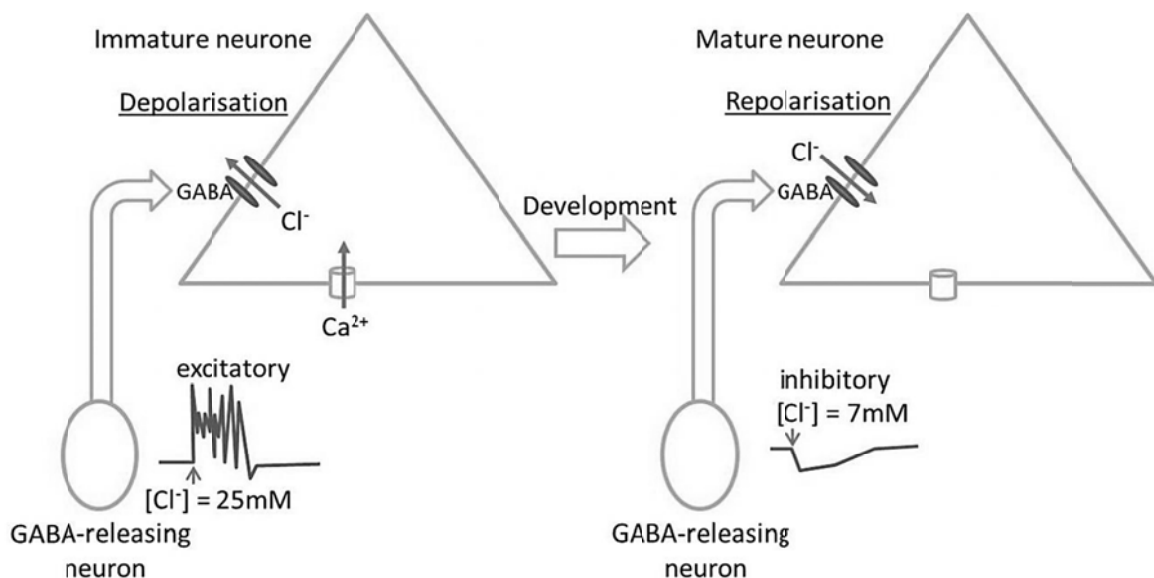


Fig. 4.2

- (c) Describe and explain how GABA is released from the pre-synaptic neurone to the post-synaptic neurone to result in an inhibitory post-synaptic potential.

1. *Depolarization of/ arrival of A.P. at pre-synaptic knob causes opening of voltage-gated Ca^{2+} channels $\rightarrow \text{Ca}^{2+}$ influx;;*
2. *Induces migration of synaptic vesicles containing GABA to CSM and fuse $\rightarrow$ release GABA into synaptic cleft via exocytosis;;*
3. *GABA diffuses across synaptic cleft (down conc grad) and binds to ligand-gated Cl^- channels, causing them to open;;*
4. *influx of Cl^- occurs $\rightarrow$ hyperpolarized of post-synaptic memb $\rightarrow$ IPSP;;*

[4]

- (d) With reference to Fig. 4.2, explain how GABA could have differing effects on the post-synaptic membrane in different neurons.

1. *25mM in immature neurone higher than that in ECF vs 7mM in mature neurone lower than that in ECF;;*
2. *Binding of GABA to ligand-gated Cl^- channels $\rightarrow$ result in Cl^- efflux in immature neurone $\rightarrow$ leads to memb depolarization = EPSP;;*
3. *but Cl^- influx in mature neurone $\rightarrow$ leads to memb hyperpolarization = IPSP;;*

Accept "IPSP" / "EPSP"

[3]

[Total: 14]

- 5 Nail-patella syndrome (NPS) is a genetic disorder that results in small, poorly developed nails and kneecaps, but can also affect many other areas of the body, such as the elbows, chest, and hips. The genetic linkage between the NPS and ABO blood group loci was the third to be identified in humans.

The NPS gene codes for a transcription factor and a gain-of-function mutation in one of its alleles leads to NPS.

An unaffected man with blood group O marries an affected female with blood group B. Fig. 5 shows the resultant pedigree.

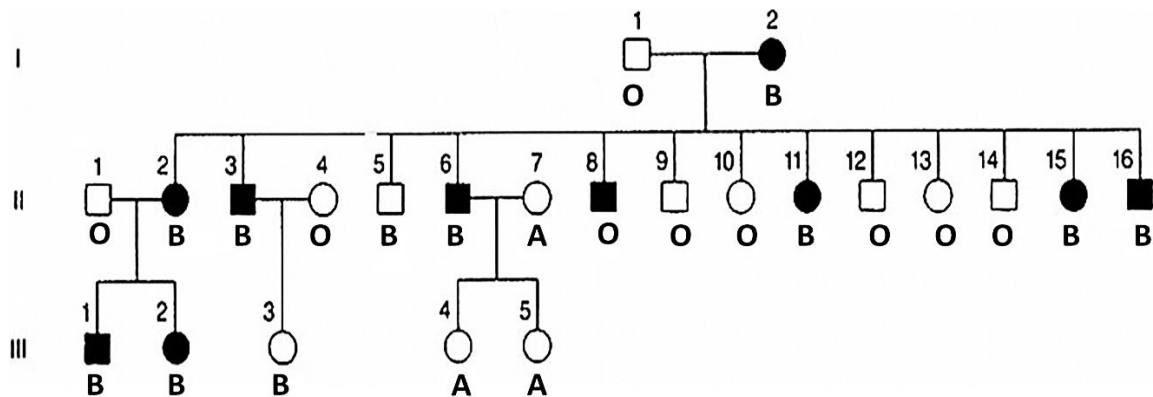


Fig. 5

(a) With reference to Fig. 5,

- (i) State and explain the mode of inheritance of NPS with respect to the ABO locus.

1. autosomal linkage between NPS and ABO loci, mutated NPS allele is dominant;;

2. all individuals with blood group B except individuals II-5 and III-3 are affected;;

3. mutant NPS allele is linked to $I^B \rightarrow$ tends to be inherited together;; [3]

- (ii) Explain why the mating between individual 6 and 7 in generation II produces 100% unaffected offsprings.

1. Genotype of 6 is $NnI^B I^O$, 7 could be $nnI^A I^A$ / $nnI^A I^O$;;

2. Offsprings inherited the I^A allele from the ♀ parent and the I^O allele from the ♂ parent;;

3. I^O allele is not linked to the mutated NPS allele;;

4. (Idea) of random segregation, independent assortment and random fertilisation;; [4]

- (b) (i) State the genotype of the individuals 5 and 8 in Generation II with respect to both the NPS and ABO loci.

1. Individual 5 $nnI^B I^O$ with both copies of the normal allele for NPS;;

2. Individual 8 $NnI^O I^O$ with one copy of the normal allele and one mutated copy;; [2]

- (ii) Suggest reasons for your answers in (i).

1. Recombinants formed from crossing over between the homologous chromosomes in prophase I;;

2. I^B allele and the mutated NPS allele separated due to chiasmata formation between their loci;

3. Some gametes contain the chromosome where the mutated NPS allele is linked to the I^O allele (accept reverse argument);; [3]

- (c) Apart from the factors due to the inheritance of genes, environment plays a very important role in determining phenotype of certain organisms.

Explain, with one example, how the environment may affect the phenotype.

1. Stating of a specific e.g. and the environmental conditions;;

2. How the phenotype differs with same genotype;;

3. Functionality of the difference in phenotype;;

[3]

[Total: 15]

- 6 Cranes are large birds and one of the endangered species, *Grus americana* commonly known as the Whooping crane is the tallest North American bird which faced near extinction due to unregulated hunting and loss of habitat.

Cranes can be categorised based on their behaviours and one of the earliest methods of classifying cranes was based on the calls they make during the breeding season.

- (a) Suggest why biologists could use calls to investigate relationships between different species of crane.

1. Calls are species specific, only members of the same species will be able to recognise the calls made;;

2. Characteristics of notes in the calls could be used to differentiate between different species;;

3. Greater the similarity in the calls, the closer the relationships between the different species;; [3]

More recently, biologists have used DNA hybridisation to confirm the relationships between different species of crane. They made samples of hybrid DNA from the same and from different species. They measured the percentage of hybridisation of each sample. The results are shown in Table 6.

Table 6

Species of crane from which hybrid DNA was made			Percentage DNA hybridisation/%
<i>Grus americana</i>	and	<i>Grus monachus</i>	97.4
<i>Grus monachus</i>	and	<i>Grus rubicunda</i>	95.7
<i>Grus americana</i>	and	<i>Grus rubicunda</i>	35.5
<i>Grus rubicunda</i>	and	<i>Grus rubicunda</i>	99.9
<i>Grus americana</i>	and	<i>Grus americana</i>	99.9
<i>Grus monachus</i>	and	<i>Grus monachus</i>	99.8

- (b) Describe the structure of the hybrid DNA between the same species.

1. 2 polynucleotides held in a double helix;;

2. held by H-bond between complementary DNA bases, A=T, C≡G;;

3. 2nm diameter, 3.4nm per complete turn;;

4. Nucleotides held by phosphodiester bonds btw phosphate grp at 5' C and 3'OH of adjacent nucleotide;;

5. *anti//, 3' → 5', 5' → 3';;*

[4]

- (c) With reference to Table 6, which of the two species seemed to be the most closely related? Explain your answer.

1. *Grus americana and Grus monachus;;*

2. *% DNA hybridisation at 97.4%, indicating most number of bases able to form complementary base pairs;;*

[2]

- (d) The biologists measured the temperatures at which the samples of hybrid DNA separated into single strands.

Explain why these temperatures could be used to find the percentage of DNA hybridisation.

1. *Higher the temp, ↑K.E required to separate DNA strands, means higher the degree of relatedness/ hybridisation;;*

2. *Many regions able to form H-bonds between complementary base pairs;;*

[2]

[Total: 11]

Section B
Answer **one** question

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections (a), (b) etc., as indicated in the question.

- 7 (a) Describe the eukaryotic processing of pre-mRNA in terms of intron splicing, polyadenylation and 5'capping. [8]

1. *Introns are non-coding regions within a gene, hence pre-mRNA introns will be spliced out via spliceosomes;;*
2. *Constitutive splicing removes all introns and joins together all exons to form one polypeptide from a gene;;*
3. *Alternative splicing joins together different combinations of exons to form different types of polypeptides from a single gene;;*
4. *Spliceosome recognises highly conserved regions/ splice sites between the 3' end of the exon and 5' end of the intron and cleave the phosphodiester bond between the nucleotides;;*
5. *After the synthesis of the polyadenylation signal, (AAUAAA), terminal transferase will bind to the 3' end of the pre-mRNA;;*
6. *Attach a long chain of nucleotides with the base adenine, forming the poly(A) tail which is not encoded in the genome;;*
7. *5' end of the pre-mRNA will be capped by the addition of a 5' modified guanine nucleotide;;*
8. *Formation of mature mRNA which will be protected from enzymatic degradation and exported out of nucleus for translation;;*

- (b) Define control elements and explain how they influence transcription in the eukaryotic genome. [6]

1. *Control elements consist of cis-acting elements which are on the same chromosome and trans-acting elements which are on different chromosomes as the gene to be regulated;;*
2. *Cis-acting elements includes promoters – core and proximal, enhancers and silencers;;*
3. *Trans-acting elements includes genes coding for basal and specific TFs – activators and repressors;;*
4. *Core promoters are adjacent DNA sequences where the basal TF and RNA pol will bind to initiate transcription;;*
5. *Proximal promoters, which are DNA sequences that are close to the core promoter allow the binding of specific TFs to increase transcription;;*
6. *Enhancers allow the binding of activators to recruit DNA bending protein so that the enhancer is brought nearer to the core promoter to stabilise the TIC, effectively increasing the rate of transcription/ amount of mRNA produced/time;;*
7. *Silencers allow the binding of repressors to result in a decrease*

in transcription rate, effectively reducing the rate of transcription/ amount of mRNA produced/time;;

- (c) State the various ways in which gene expression of an enzyme may be controlled at translational and post-translational level. [6]

Translational level

1. *mRNA stability by having a long poly(A) tail and 5' cap, shortening of poly (A) tail & removal of cap results in the recruitment of nucleases to degrade the mRNA;;*
2. *Gene silencing when microRNA associates with the RISC (spell in full);;*
3. *results in either degradation of the target mRNA, or the translation inhibition of the target mRNA;;*

Post-translational level

4. *Concentration/ stability of the translated product could be regulated via the degradation of the protein;;*
5. *Proteins designated for destruction are conjugated with a small protein called ubiquitin;;*
6. *Ubiquitinated proteins are subsequently recognized and degraded by proteasome;;*
7. *GA modification of the translated polypeptide by cleaving the polypeptide or modifying the sugar monomers on the oligosaccharide chains/ phosphorylation;;*

[Total: 20]

- 8 (a) Describe the role of mitosis in maintaining genetic stability. [4]
1. *formation of two nuclei, with same no. of chromosomes as the parent cell;;*
 2. *During replication, each strand acts as template for synthesis of new daughter strands;;*
 3. *via complementary bp, A-T 2H-bonds, CG 3H-bonds;;*
 4. *=separation of sister chromatids at anaphase ensure = distribution of genetic materials to both daughter cells;*
 5. *genetically identical daughter cells with no variation;;*
 6. *production of clones during asexual reproduction to colonise;;*
- (b) Explain how meiosis can lead to genetic variation. [8]
1. *diagrams showing prophase I, metaphase I, anaphase I, metaphase II, anaphase II*
 2. *independent assortment of homologous chromosomes at metaphase plate in anaphase I;*
 3. *possible gametic combinations, 2ⁿ*
 4. *crossing over at chiasmata during prophase I between non-sister chromatids of homologues;;*
 5. *linked genes separated at chiasmata, formed new combinations of alleles;;*
 6. *sister chromatids no longer identical, will segregate randomly*

during anaphase II;;

7. *new gametes produced from parents not identical, with only one of each pair of chromosomes;;*
8. *recombinant genotypes gave rise to genetic variation;;*
9. *un= separation due to erroneous spindle fibre formation leads to polyploidy if there are no separation of sis chromatids;*
10. *Aneuploidy may result if there are un= separation in one pair of sister chromatids;;*

(c) With one named example, describe the role of proto-oncogenes and tumour suppressor genes in the control of normal cellular division. [8]

1. *proto-oncogene = genes that code for proteins that stimulate normal cell growth and division.*
2. *RAS, codes for a G prot, role in G1 check point where the presence of growth factor is being checked for;;*
3. *Binding of growth factor result in phosphorylation cascade to result in the increase in levels of cyclins in cell;;*
4. *Cyclins bind to CDKs, forming complexes to remove the inhibition of other proteins that prevents progress to S phase;;*
5. *In the absence of growth factor, the RAS is not activated, hence cells do not progress to S phase/ presence of growth factor, Ras activated → cells progress from G1 → S phase;;*
6. *p53, tumour suppressor genes = code for proteins that normally inhibit cell division (when conditions for cell division not met);;*
7. *by activating DNA repair proteins to repair damaged DNA / controlling proper cell anchorage;;*
8. *activating transcription of p21 gene → p21 binds to Cdks → prevent entry into S phase;;*
9. *triggering apoptosis (when DNA damage is beyond repair);;*

[Total: 20]

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